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Convergence in distribution of fractional Gaussian Fields

Now that we have presented and proved the existence of fractional Gaussian fields X_s on K and discrete fractional Gaussian fields X_s^m on V_m , our next task is proving that they converge. Therefore, in this section our main goal is to show that X_s^m converges in distribution to X_s .

1.1 Preliminary Lemmas

In this section we will prove a collection of lemmas that will be needed.

Lemma 1.1: Let $\{P_t\}_{t \geq 0}$ be the semigroup generated by the Dirichlet Laplacian $(-\Delta)^{-s}$ and $p_t(x, y)$ the heat kernel admitted by $\{P_t\}_{t \geq 0}$. If ϕ_i is an orthonormal eigenfunction of $(-\Delta)^{-s}$ then

$$P_t \phi_i(x) = e^{-\lambda_i t} \phi_i(x).$$

Proof. The lemma follows immediately from the calculations:

$$\begin{aligned} P_t \phi_i(x) &= \int_K p_t(x, y) \phi_i(y) \, d\mu(y) = \int_K \left(\sum_{j=1}^{\infty} e^{-\lambda_j t} \phi_j(x) \phi_j(y) \right) \phi_i(y) \, d\mu(y) \\ &= \int_K e^{-\lambda_i t} \phi_i(x) \, d\mu(y) = e^{-\lambda_i t} \phi_i(x) \end{aligned}$$

□

Lemma 1.2: For all $f \in S(K)$ and $t > 0$,

$$\lim_{m \rightarrow \infty} \frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m f_m(p) = \int_K f(x) P_t f(x) \, d\mu(x)$$

where $f_m = f|_{V_m}$.

Proof. In this proof we wish to use **KURTZ1969354** so we will check that Δ fulfils the criteria. First there is a bounded linear operator such that for all $f \in L^2(K, \mu)$ we have

$$\lim_{n \rightarrow \infty} \|P_n f\| = \|P_n\|.$$

This holds in our case, since we have $P_n f = f|_{V_n} = f_n$.

Secondly we note that Δ is the infinitesimal generator of a uniformly continuous semigroup by ??.

Next we have that the sequence of $\{\Delta_m\}_{m \geq 0}$ indeed coincides with the extended limit defined, in **KURTZ1969354**, this follows from ?? and ??.

Finally the range $\mathcal{R}(\lambda_0 - \Delta)$ is dense in L^2 by **fabriceDirichelt**. Thus by **KURTZ1969354** we have that for all $t \geq 0$

$$\lim_{m \rightarrow \infty} \sup_{p \in V_m} |P_t^m f_m(p) - (P_t f(p))_m| = 0.$$

If we now write

$$\frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m f_m(p) = \int_{V_m} f_m P_t^m f_m \, d\mu_m$$

then

$$\int_{V_m} f_m P_t^m f_m \, d\mu_m = \int_{V_m} f_m (P_t^m f_m - (P_t f)_m) \, d\mu_m + \int_{V_m} f_m (P_t f)_m \, d\mu_m.$$

The first integral converges to 0 as $m \rightarrow \infty$ because of (1.1). Further, by ?? we have $\int_{V_m} f_m (P_t f)_m \, d\mu_m \rightarrow \int_K f P_t f \, d\mu$ as $m \rightarrow \infty$. $\square$

Lemma 1.3: For all $f \in S(K)$ and $s \geq 0$ the following holds

$$\frac{1}{a_m} \sum_{p \in V_m} f_m(p) (-\Delta_m)^{-2s} f_m(p) \rightarrow \int_K f(x) (-\Delta)^{-2s} f(x) \, d\mu(x), \quad \text{as } m \rightarrow \infty.$$

Proof. For $s = 0$ we find that

$$\begin{aligned} (-\Delta_m)^{-0} f &= \sum_{i=1}^{N_m} \frac{1}{(\lambda_i^m)^0} \phi_i^m(x) \frac{1}{a_m} \sum_{y \in V_m} \phi_i^m(y) f(y) \\ &= \sum_{i=1}^{N_m} \phi_i^m(x) \int_{V_m} \phi_i^m(y) f(y) \, d\mu_m \\ &= f_m. \end{aligned}$$

Likewise by ?? we find

$$(-\Delta)^{-0} f = \sum_{i=1}^{\infty} \phi_i \int_K \phi_i(y) f(y) \, d\mu = f$$

and the lemma for $s = 0$ follows from ??.

Now assume $s > 0$, then we first note that

$$\begin{aligned} (-\Delta_m)^s f_m &= \frac{1}{a_m} \sum_{j=1}^{N_m} (\lambda_j^m)^{-s} \phi_j^m(x) \sum_{y \in V_m} \phi_j(y) f_m(y) \\ &= \frac{1}{a_m} \sum_{j=1}^{N_m} \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-\lambda_j t} dt \phi_j^m(x) \sum_{y \in V_m} \phi_j(y) f_m(y) \\ &= \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} \frac{1}{a_m} \sum_{y \in V_m} \sum_{j=1}^{N_m} e^{-\lambda_j t} \phi_j^m(x) \phi_j(y) f_m(y) dy. \end{aligned}$$

Now by Fubini's theorem and ?? we have that

$$\frac{1}{a_m} \sum_{p \in V_m} f_m(p) (-\Delta_m)^{-2s} f_m(p) = \frac{1}{\Gamma(2s)} \int_0^\infty t^{2s-1} \frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m f_m(p) dt.$$

By Lemma 1.1 we find that

$$\begin{aligned} \frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m f_m(p) &= \frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m \sum_{j=1}^\infty \langle f_m, \phi_j \rangle \phi_j \\ &= \frac{1}{a_m} \sum_{p \in V_m} f_m(p) \sum_{j=1}^\infty \langle f_m, \phi_j \rangle P_t^m \phi_j \\ &= \frac{1}{a_m} \sum_{p \in V_m} f_m(p) \sum_{j=1}^\infty \langle f_m, \phi_j \rangle e^{-\lambda_j^m t} \phi_j \\ &\leq e^{-\lambda_1^m t} \frac{1}{a_m} \sum_{p \in V_m} f_m(p)^2 \end{aligned}$$

where we can guarantee the inequality for all j since $0 < \lambda_1 < \lambda_2 < \dots$

Note that $\sup_m \frac{1}{a_m} \sum_{p \in V_m} f_m(p)^2 < \infty$ since $f \in S(K) \subset C(K)$ whereby f is bounded and $\#V_m < \infty$ for all m . We can now by dominated convergence, Fubini's theorem and Lemma 1.2 find that

$$\begin{aligned} \lim_{m \rightarrow \infty} \frac{1}{a_m} \sum_{p \in V_m} f_m(p) (-\Delta_m)^{-2s} f_m(p) &= \frac{1}{\Gamma(2s)} \int_0^\infty t^{2s-1} \lim_{m \rightarrow \infty} \frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m f_m(p) dt \\ &= \frac{1}{\Gamma(2s)} \int_0^\infty t^{2s-1} \int_K f(x) P_t f(x) d\mu(x) dt \\ &= \frac{1}{\Gamma(2s)} \int_K f(x) \int_0^\infty t^{2s-1} P_t f(x) dt d\mu(x) \\ &= \int_K f(x) (-\Delta)^{-2s} f(x) d\mu(x) \end{aligned}$$

as desired. $\square$

1.2 Main Theorem

With everything in mind we can now prove the convergence in distribution of discrete fractional Gaussian fields to fractional Gaussian fields.

Theorem 1.4: Let $s \geq 0$, then X_s^m converges to X_s in distribution in $S'(K)$ when $m \rightarrow \infty$

Proof. Since $S(K)$ is a nuclear space, by **biernÅl2017generalizedrandomfieldslevys** it is enough to prove the convergence of the characteristic functional, e.i that for all $f \in S(K)$

$$\mathbb{E} \left[\exp \left(it X_s^m(f_m) \right) \right] \rightarrow \mathbb{E} \left[\exp \left(it X_s(f) \right) \right], \quad \text{as } m \rightarrow \infty$$

where $f_m = f|_{V_m}$. It follows from ?? that

$$\mathbb{E} \left[\exp \left(it X_s^m(f_m) \right) \right] = \exp \left(-\frac{t^2}{2} \mathbb{E} \left[(X_s^m(f_m))^2 \right] \right) = \exp \left(-\frac{t^2}{2a_m} \sum_{p \in V_m} f_m(p) (-\Delta_m)^{-2s} f_m(p) \right).$$

Likewise, via ?? we find

$$\mathbb{E} \left[\exp(it X_s(f)) \right] = \exp \left(-\frac{t^2}{2} \mathbb{E} \left[(X_s(f))^2 \right] \right) = \exp \left(-\frac{t^2}{2} \int_K f(-\Delta)^{-2s} f \, d\mu \right)$$

where we notice by ?? that

$$\begin{aligned} \mathbb{E} \left[(X_s(f))^2 \right] &= \int_K (-\Delta)^{-s} f (-\Delta)^{-s} f \, d\mu \\ &= \int_K \int_K f(p) f(q) G_{2s}(p, q) \, d\mu(p) \, d\mu(q) \\ &= \int_K f(-\Delta)^{-2s} f \, d\mu. \end{aligned}$$

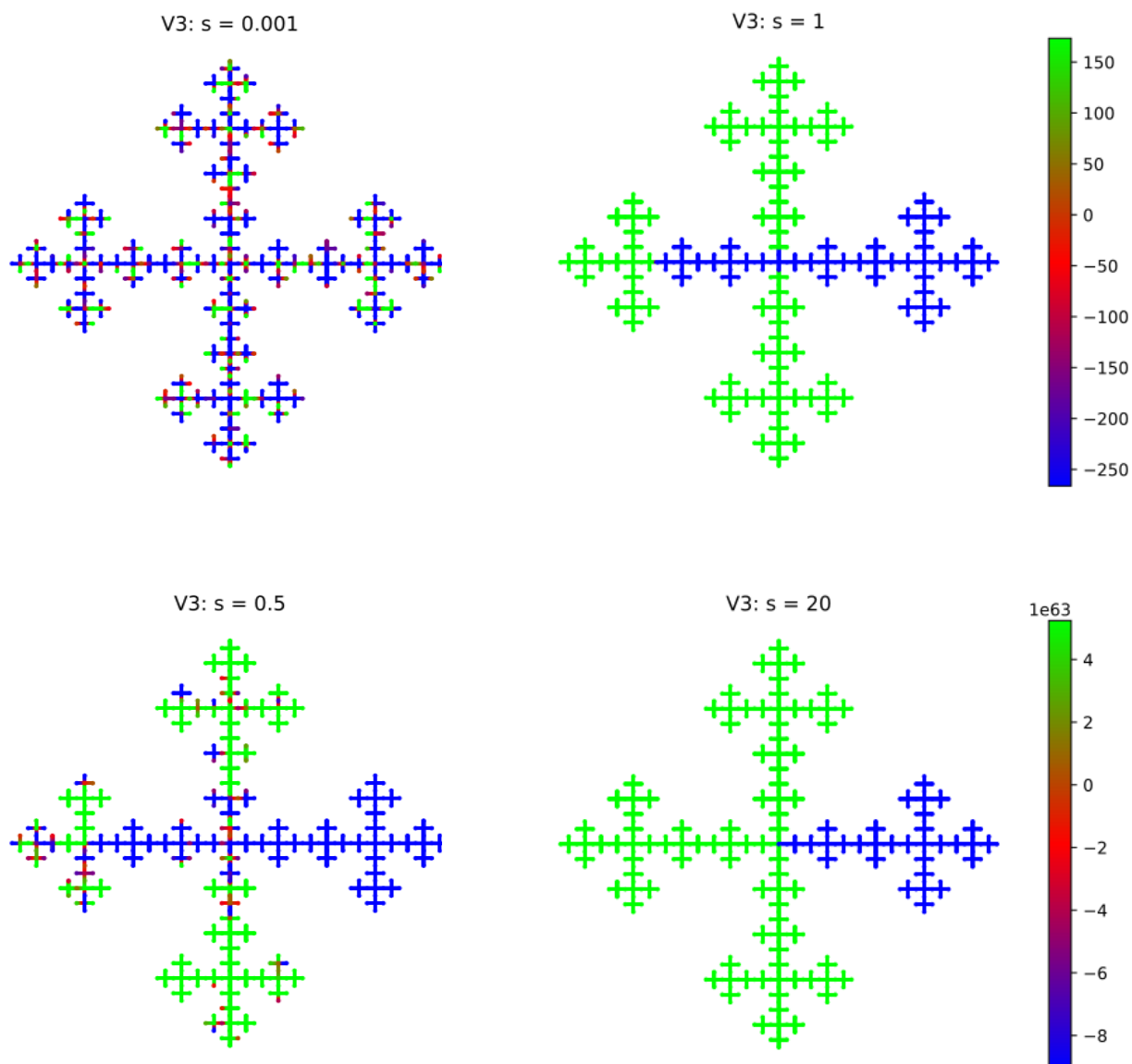
The proof now follows from lemma 1.3. □

Appendix A - Simulation Images

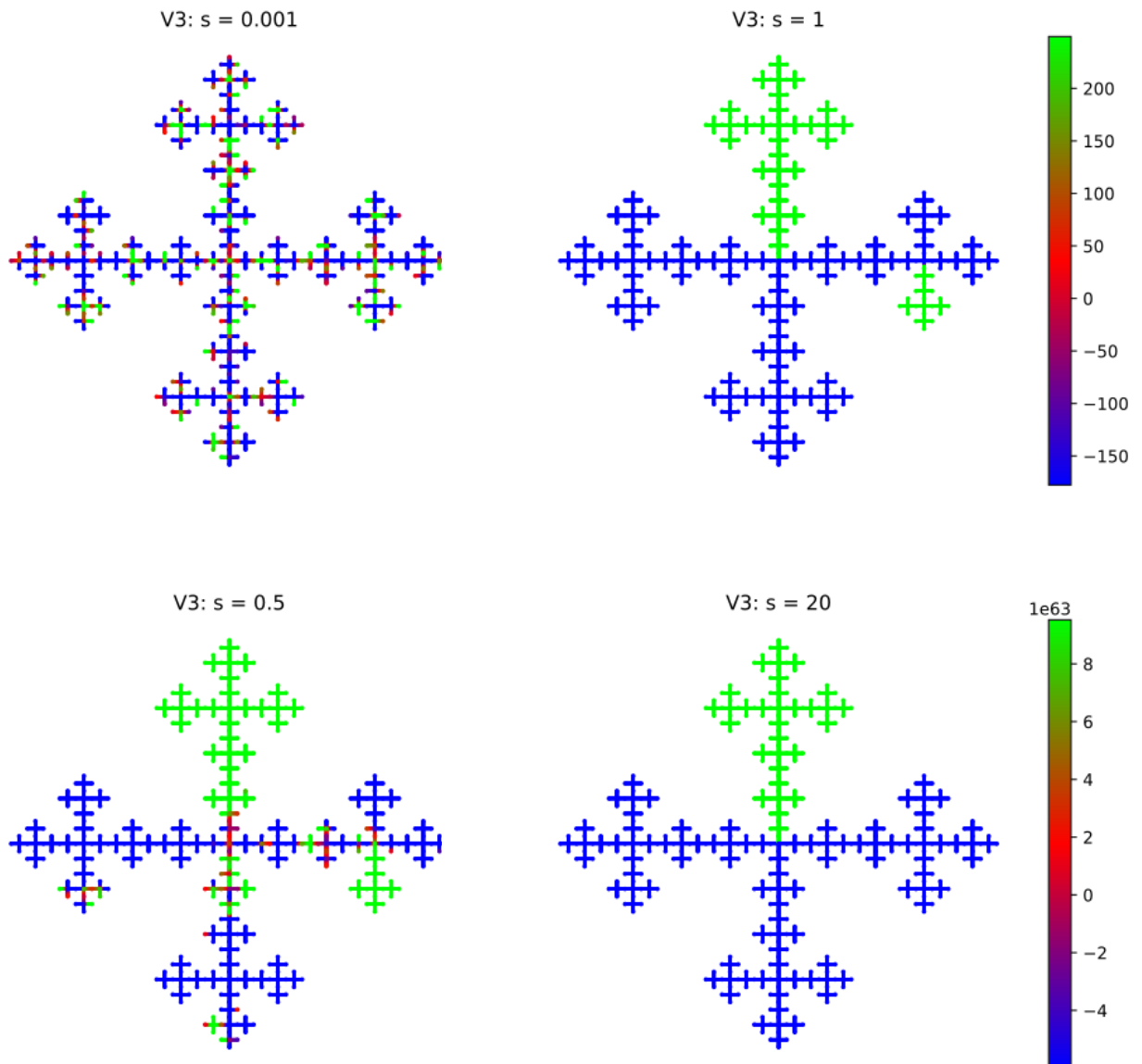
In this appendix we have included larger version of the images from ??.

Full scale image of ??

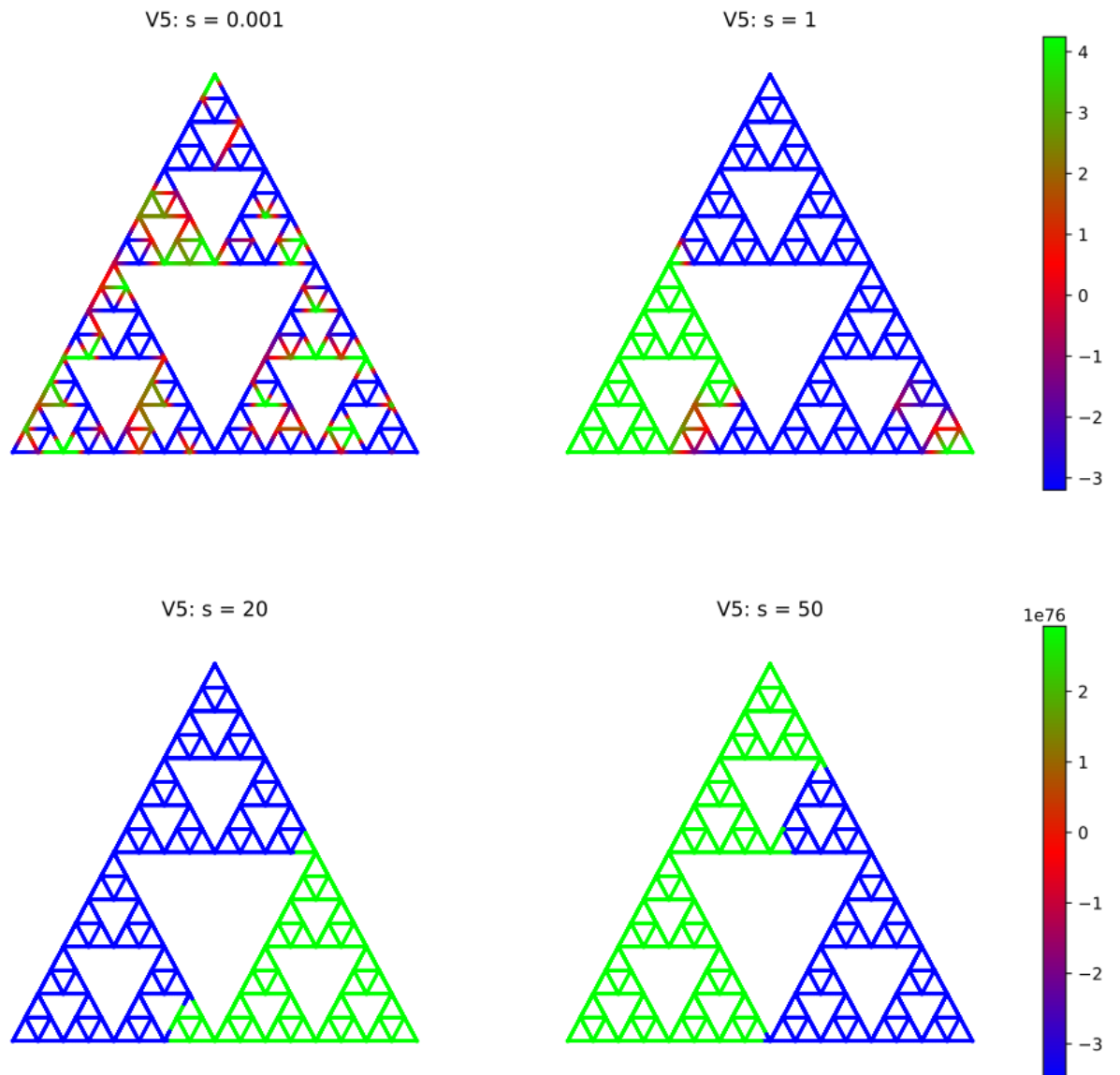
Full scale image of ??



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